

XIAOYING LIAO

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INTRODUCTION

I am not afraid of challenges, and I am determined to pursue my goals. I dream of becoming a biological scientist, exploring the unknowns of life, and making meaningful discoveries. I am deeply interested in both biology and artificial intelligence, and I hope to bring them together to help us understand biological systems in new ways.

Personal website: <https://cristal0412.github.io/>

EDUCATION

Johns Hopkins University (JHU)

Aug. 2024 – May 2025

Master of Science in Engineering (M.S.E.) in Biomedical Engineering

Baltimore, USA

- Supervised by *Professor Carlo Iomini*, **GPA: 4.00/4.00**

- Courses:** Cellular Engineering, Tissue Engineering, Cell & Tissue Engineering Lab, Advanced Topics in Regenerative and Immune Engineering, Human Genetics, Systems Pharmacology and Personalized Medicine

University of Sydney (USYD)

Feb. 2020 – Dec. 2023

Bachelor of Science (Honours) in Statistics

Sydney, Australia

- Supervised by *Professor Jean Yang*, **GPA: 3.90/4.00**, **Rank: 1/200**

- Courses:** Probability and Estimation Theory, Statistical Inference, Stochastic Process, Advanced Statistical Modelling, Statistical Machine Learning, Life and Evolution

PUBLICATIONS

Python Song, **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026. **Oral (Top 1%)**

Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 2026. pp. 3883.

Huanhuan Xiao, Xiangqian Shi, Abi Cha, Emma Liu, Randi Marshall, **Xiaoying Liao**, Molly Voit, Uzoamaka Nwagbo, Frazer Matthews, James H. Thierer, Meera Saxena, Ximin Li, Tae-In Kam, Xiangrong Kong, Jeff S. Mumm, Liyun Zhang. *Novel zebrafish genetic models of pde6b mutants exhibit rod photoreceptor degeneration*. Manuscript completed; planned submission to *Nature Communications Biology* by October 2026.

Nianping Liu, Xuanzhi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.

Huanhuan Xiao, Liyun Zhang, **Xiaoying Liao**, Emma Liu, Shaojun Du. *Expression and functional analysis of Hspa8 in early zebrafish development*. Manuscript completed; planned submission to *Cells* in 2026.

Tianyang Wang, Xi Xiao, **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026. Under review.

Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.

Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smrita Krishnaswamy, Jun Huan, Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.

Tianyang Wang, Xi Xiao, Gaoqi Chen, **Xiaoying Liao**, Guo Cheng, Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025. pp. 1124–1131.

Xiaoying Liao. *Assessing the Performance of Clustering Methods in Spatially Resolved Transcriptomics Data*. University of Sydney, 2023. Undergraduate honors research.

RESEARCH EXPERIENCE

Stanford University

2026–Present

Research Assistant, supervised by Prof. Xiaojie Qiu

Stanford, CA

- Human fetal heart development:** Analyzing PCW7 and 8 human fetal heart spatial transcriptomics data to characterize cardiac cell states, spatial niches, cell–cell communication, and 3D reconstruction during early heart development.
- 3D embryonic development:** Building a 3D spatiotemporal atlas of mouse embryogenesis from E5.5–E7.5 to characterize cell-state transitions, spatial tissue organization, and developmental trajectories during early embryogenesis.
- Pediatric cardiomyopathy:** Leading a single-cell multi-omics and spatial transcriptomics study comparing myocardial states before and after ventricular assist device support (VAD) and between recovery and non-recovery to identify cellular programs, spatial niches, and potential biomarkers associated with myocardial recovery.

Johns Hopkins University

2024–2026

Research Assistant, supervised by Prof. Carlo Iomini

Baltimore, MD

- **Meibomian gland development:** Investigated how Hedgehog-responsive epithelial progenitors regulate gland development, meibocyte maturation, and lipid homeostasis.
- **Cilia–Hedgehog signaling:** Characterized Ift88–Smo genetic interactions and their effects on epithelial proliferation, gland morphology, and lipid production and composition.
- **Single-cell transcriptomics:** Generated and analyzed P5 and P10 meibomian gland scRNA-seq data to characterize developmental cell states and pathways associated with gland maturation and Hedgehog signaling.

University of Sydney

2023–2024

Research Assistant, supervised by Prof. Jean Yang and Prof. Clara Grazian

Sydney, Australia

- **Data preprocessing:** Used Python and R to preprocess spatial transcriptomics datasets, including quality control, normalization, feature selection, and dimensionality reduction using principal component analysis (PCA).
- **Spatial domain detection:** Applied and compared clustering approaches, including K-means and Seurat, alongside spatially informed methods such as BayesSpace and STitch3D, to identify tissue domains and characterize spatial organization.
- **Method evaluation:** Benchmarked clustering performance across simulated and real datasets, examining factors affecting method efficacy and identifying strengths and limitations to inform method selection in spatial transcriptomics research.

TEACHING EXPERIENCE

Johns Hopkins University

Jan. 2025 – May 2026

Teaching Assistant

Baltimore, United States

- **Cell & Tissue Engineering Lab:** Mentored students in practical modules on cell culture, gene delivery, metabolic glycoengineering, and tissue modeling. Provided support in data collection, data analysis, and scientific lab reports.

University of Sydney

Feb. 2023 – Dec. 2023

Teaching Assistant

Sydney, Australia

- **STAT2011 Probability and Estimation Theory:** Guided students through probability theories and statistical distributions, including practical R programming for data analysis applications.
- **DATA2002 Data Analytics: Learning from Data:** Led workshops on data analytics techniques, enabling students to apply statistical programming to real-world data challenges and machine learning concepts for informed decision-making.

PROFESSIONAL EXPERIENCE

JD.com

Feb. 2022 – May 2022

Data Analyst Intern

Beijing, China

- **Retail Dataset Management:** Utilized Hive SQL for complicated retail sales data management, focusing on home appliances at JD.com. Implemented time-series forecasting and multivariate regression to improve inventory and promotion strategies support for millions of users.
- **Consumer Behavior Analytics:** Analyzed consumer behavior using advanced statistical modeling in R and Python to identify key purchasing patterns. Developed and implemented sales strategies tailored to customer segments, resulting in a 25% increase in sales performance compared to the previous quarter.
- **Operational Analytics Platform Development:** Designed and built an advanced analytics dashboard using Python, enabling the operations team to track daily sales data in real time. Facilitated the rapid development of targeted business strategies, which led to a quantifiable 60% improvement in operational efficiency.

ByteDance – TikTok

Dec. 2021 – Feb. 2022

Data Operation Intern

Beijing, China

- **Product Optimization:** Performed market research for CRM platforms, delivering bespoke solutions to meet diverse global sales and operational challenges.
- **Data Empowerment:** Monitored business performance through SQL and Python, automatically analyzed vital indicators, and provided timely feedback on potential problems. Established an interactive BI data dashboard to display key business indicators visually and support real-time decision-making.
- **Brand Growth:** Conducted data analysis of user behavior to develop an AIPL (Awareness, Interest, Preference, and Loyalty) user classification model. Tailored payment tools, strategies, and creative content to users at various stages of brand awareness, achieving a remarkable 15% increase in brand advertising click-through rate.

ACADEMIC ACTIVITIES

Stanford Genetics Department Retreat: Poster presentation

2026

Association for Research in Vision and Ophthalmology (ARVO): Poster presentation

2026

Johns Hopkins Cell Biology Symposium: Poster presentation

2026

Annual Wilmer Research Meeting: Poster presentation

2025

Sydney Bioinformatics Research Symposium (SBRs): Participant

2023

Unlocking Single Cell Spatial Omics Analyses with scDney: Participant

2023

SKILLS & HONORS

Computational: Python, R, SQL, MATLAB, PyTorch, scikit-learn, Scanpy, Seurat; machine learning, statistical modeling, single-cell and spatial transcriptomics, multi-omics integration

Experimental: 10x Genomics single-cell sequencing, Illumina spatial transcriptomics, cell and tissue culture, immunostaining, Confocal and two-photon microscopy, PCR/qRT-PCR, Western blotting, DNA/RNA extraction

Honors: Vice Chancellor's Global Mobility Scholarship (2020); Dalyell Scholar (2022); 2nd Prize, China National High School Biology Olympiad (2018)